

Fractions



2.1 Fractions

We will learn to convert improper fractions to numbers mixed and vice versa.

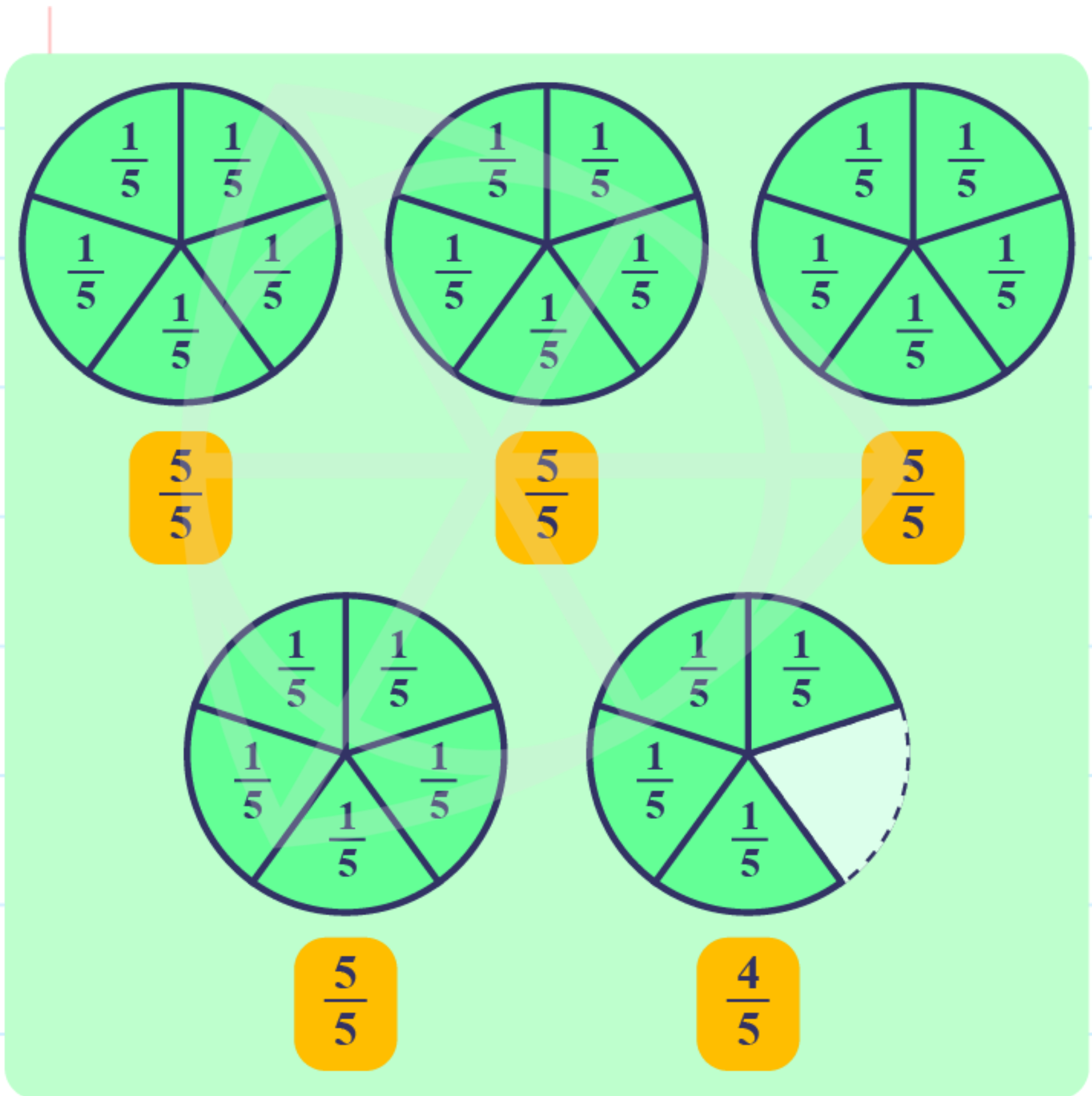
- An improper fraction is a fraction with the numerator equal to or greater than the denominator.

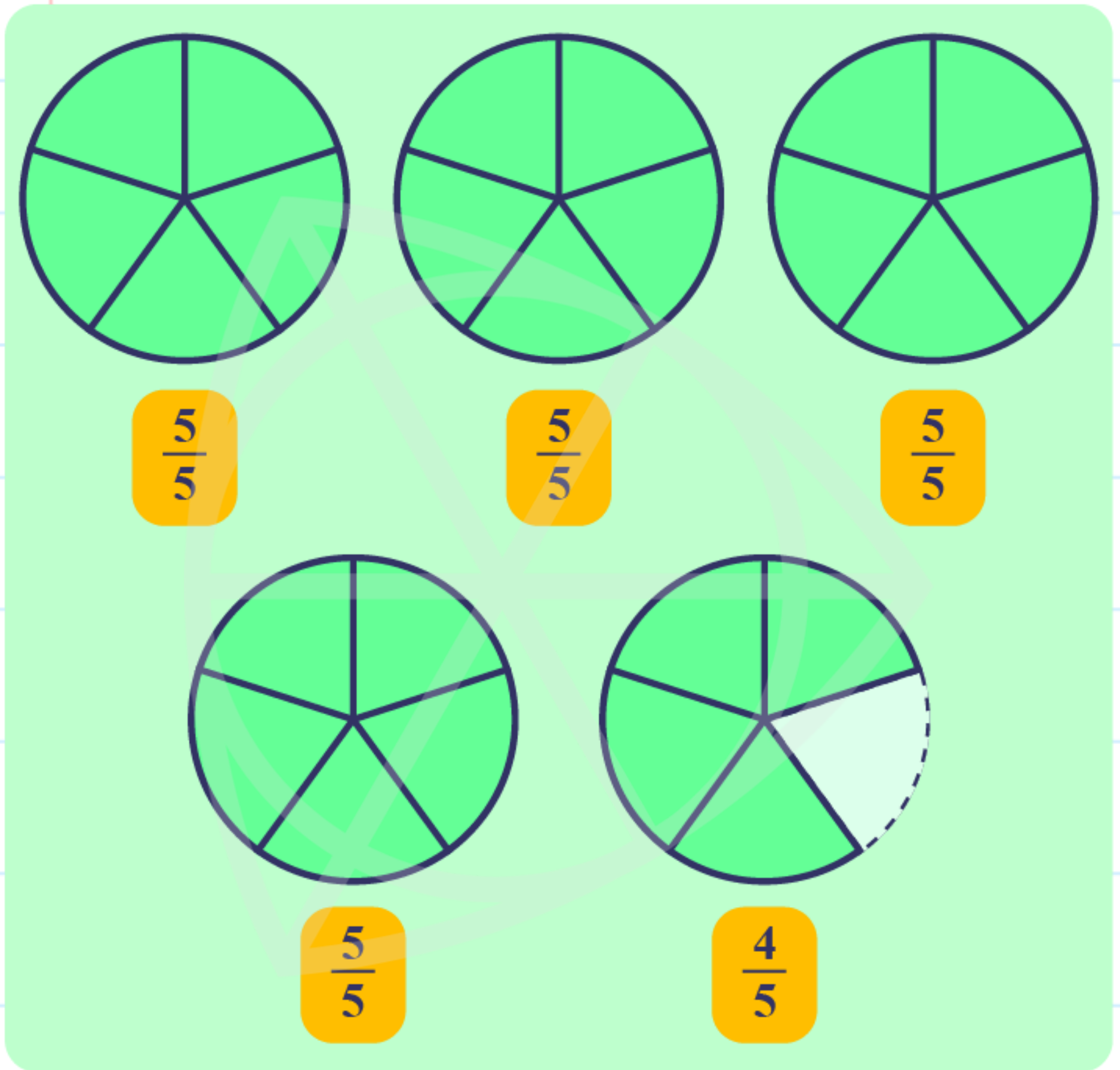
As example,

$\frac{24}{5}$ is an improper fraction because the value of the numerator (24) is greater than the value of the denominator (5).

Meanwhile, a mixed number consists of a whole number and a proper fraction.

As example, $\frac{24}{5}$ can be converted to mixed numbers as follows:





Based on the above picture, $\frac{24}{5} = 4\frac{4}{5}$.

Or, the simple steps to convert improper fractions to mixed numbers are as follows:

$$\begin{array}{r} 4 \\ 5 \overline{)24} \\ \underline{-20} \\ 4 \end{array} = 4\frac{4}{5}$$

The simple steps to convert mixed numbers to improper fractions are as follows:

$$4\frac{4}{5} = \frac{4 \times 5 + 4}{5}$$
$$= \frac{24}{5}$$

- To **add** fractions with the same denominators, add the numerators without changing the denominators.

As example,

$$\frac{1}{5} + \frac{2}{5} = \frac{1 + 2}{5}$$
$$= \frac{3}{5}$$

- To **add** fractions with different denominators, find the common multiple value of the denominator.

As example,

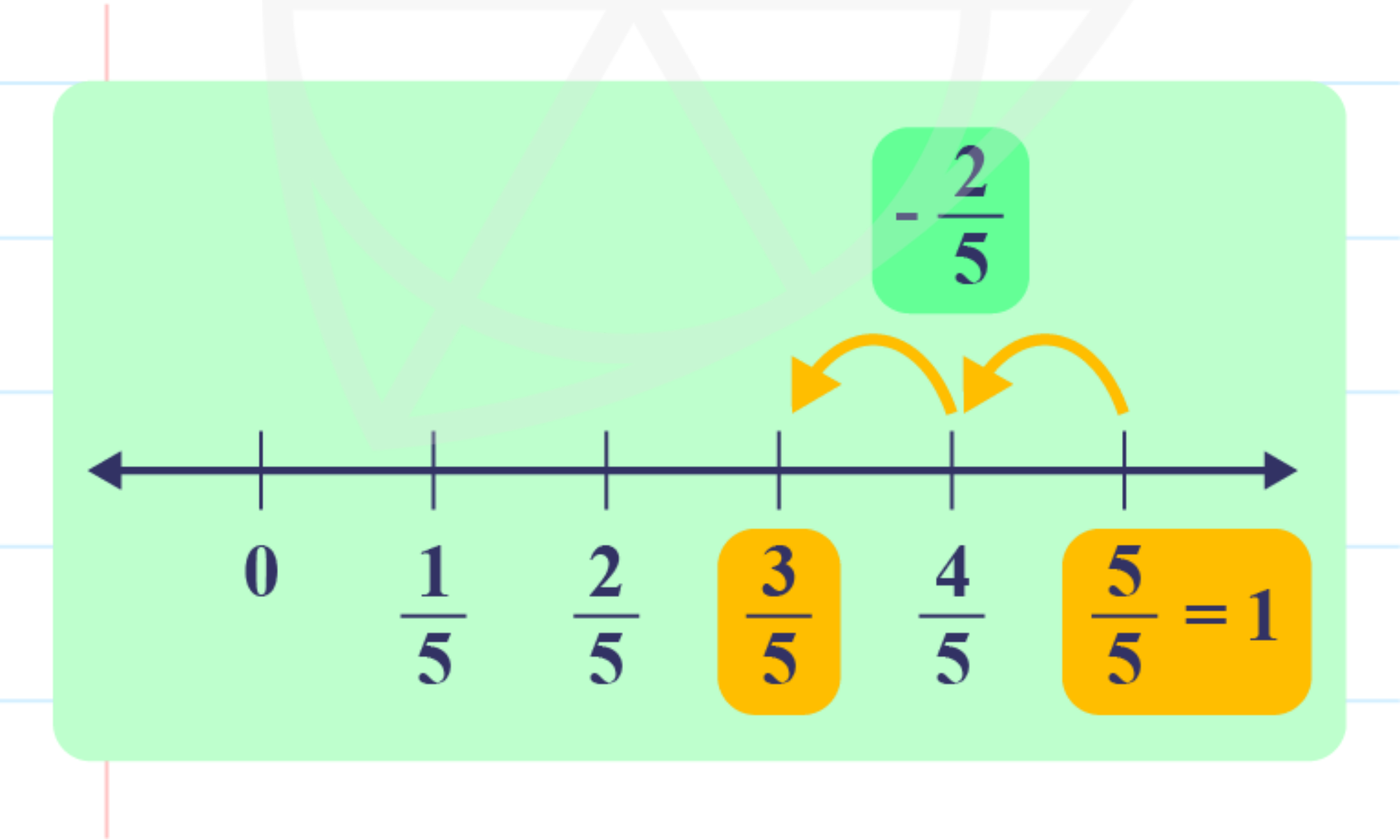
$$\frac{1}{6} + \frac{3}{4} = \frac{1 \times 2}{6 \times 2} + \frac{3 \times 3}{4 \times 3}$$
$$= \frac{2}{12} + \frac{9}{12}$$
$$= \frac{11}{12}$$

- To **subtract** fractions with the same denominator, the subtraction operation is performed as usual.

The denominator is retained.

As example,

$$1 - \frac{2}{5} = \underline{\hspace{2cm}}$$



Thus, $1 - \frac{2}{5} = \frac{3}{5}$.

To subtract the fractions with different denominators, change to equivalent fractions with the same denominators before subtracting the numerators.

Retain the denominator.

As example,

$$\begin{aligned}\frac{3}{5} - \frac{1}{2} &= \frac{3 \times 2}{5 \times 2} - \frac{1 \times 5}{2 \times 5} \\ &= \frac{6}{10} - \frac{5}{10} \\ &= \frac{1}{10}\end{aligned}$$

- For mixed operations involving the addition and subtraction of fractions, the calculation is done from the left to the right.

As example,

$$\begin{aligned}\frac{1}{7} + \frac{5}{7} - \frac{2}{7} &= \frac{1 + 5 - 2}{7} \\ &= \frac{4}{7}\end{aligned}$$

- 'from' means **multiply**

As example, find the value of $\frac{5}{7}$ from 56.

$$\begin{aligned}\frac{5}{7} \text{ from } 56 &= \frac{5}{7} \times 56 \\ &= \frac{5 \times 56}{7} \\ &= \frac{280}{7} \\ &= 40\end{aligned}$$